

CURRICULUM, ASSESSMENT AND DATA EVIDENCE

Mathematics and STEM Enrichment Programme

Public evidence sample prepared for portfolio use

Context

I contributed to a structured academic enrichment programme for a small class of 10 high-attaining students at a Chinese international school. The programme was designed for Year 9/10 equivalent learners preparing for more demanding study in mathematics, science, technology and academic reasoning.

My role included developing selected STEM and mathematics-focused sessions, creating short assessment tasks and pop quizzes, and using evidence from learner responses to reflect on understanding after each session. The original course timetable has been rewritten here for public use: school names, logos, staff names, exact dates and internal branding have been removed.

Evidence focus

Curriculum sequencing, STEM lesson development, formative assessment, pop-quiz data, academic communication and responsible use of digital learning tools.

Tools and delivery

Live delivery	VooV was used for online sessions, with expectations around camera, microphone and participation.
Learning resources	Lesson slides, structured worksheets, homework prompts and short explanation tasks.
Assessment checks	Wayground pop quizzes, quick checks, exit prompts, micro-presentations and short written responses.
Data handling	A spreadsheet-based tracker was used to summarise understanding, common errors and follow-up needs. Raw student data is not included in this public sample.

Curriculum - assessment - data - reflection

01

Programme design

The source timetable was built as a 24+ hour programme delivered through 1.5 hour sessions across the academic year. It moved from critical thinking and independent learning towards non-routine problem solving, communication and review.

Stage	Public-facing curriculum purpose	Assessment evidence gathered
1. Critical thinking	Students worked with scientific ideas, evidence, definitions and validity of conclusions.	Short reasoning prompts, precision checks and written explanations.
2. Independent learning	Students practised active reading, claim-evidence mapping, visual interpretation and academic summary.	Annotated extracts, concise summaries and visual interpretation tasks.
3. Problem solving	Students applied mathematics, science, computing and engineering ideas to unfamiliar problems.	Wayground pop quizzes, multi-step problem tasks and explanation of assumptions.
4. Communication and review	Students developed academic speaking, interview-style reasoning, presentation skills and reflection.	Micro-presentations, peer scoring, mixed quiz and reflective learning logs.

Rewritten snippets from the source timetable

Evidence evaluation	Students analysed case studies and data, weighed benefits and risks, and practised selecting the best-supported explanation.
Learning habits	Homework, session summaries, vocabulary lists and learning logs were used to keep learning visible between sessions.
Progression	The course ended with presentation, quiz and reflection tasks so that students could review reasoning, communication and next steps.

02

Selected contribution areas

The strongest evidence is not the timetable alone. It is the link between planned sequence, STEM lesson design, assessment tasks and the data collected from student responses.

Contribution	What was developed	Professional evidence shown
Precision in mathematical and scientific language	A session on accuracy, definitions, units, bounds and how small wording changes can affect a conclusion.	Clear mathematical language, misconception checking, exam-style reasoning and precision in explanation.
Creative problem-solving methods	A structured approach to non-routine problems using tools, information, proposal and final statement.	Problem decomposition, method selection, algebraic reasoning and explanation of strategy.
Computing, AI and decision-making	Activities on algorithms, logic, decision trees, bias and simple pseudo-AI systems.	Computational thinking, responsible AI discussion, structured reasoning and digital literacy.
Interview and persuasion practice	Mini-interview prompts where students thought aloud, justified claims and responded to probing questions.	Academic communication, reasoning under pressure and structured response design.
Final quiz and presentation review	A mixed quiz and peer-assessed micro-presentation task combining STEM content, reasoning and reflection.	Formative data collection, communication assessment, presentation scoring and reflective learning.

How this reads as evidence

This document shows contribution to a coherent academic programme rather than isolated one-off lesson planning. The selected sessions connect mathematics, STEM, reasoning, assessment and data-informed reflection.

03

Assessment and data collection

Short formative assessment tasks were used to collect evidence of understanding without turning each session into a formal test. The emphasis was on identifying misconceptions, confidence and next steps for teaching.

Method	Purpose	Data or evidence produced	How it informed planning
Wayground pop quiz	Check recall and immediate application after a session.	Score bands, item patterns and common wrong answers.	Identify recap points and adjust the following session's emphasis.
Exit prompt	Capture one point understood and one area of uncertainty.	Short qualitative responses.	Plan follow-up questions and improve explanation clarity.
Homework task	Assess whether students could apply ideas independently.	Written explanations, calculations or presentation preparation.	Provide targeted feedback and track development of reasoning.
Presentation / interview response	Assess clarity, structure and reasoning under time pressure.	Rubric notes on evidence, communication and logic.	Guide individual feedback and progression recommendations.
Learning log	Encourage students to record what they learnt and what they found challenging.	Five-point summaries or reflective notes.	Identify what students remember and what they consider important.

Public data handling

For public use, the evidence is deliberately summarised. It does not include raw scores, individual rows, student names, screenshots or identifiable work. Appropriate public summaries could include grouped trends such as: students improved in explaining assumptions; several students needed more support with wording and units; or quiz responses showed that multi-step reasoning required further modelling.

04

Professional reflection

This work shows the development of a learning sequence, not simply delivery of a single activity. The programme linked subject knowledge with academic reasoning, then used short assessment moments to make understanding visible.

Professional practice	Evidence demonstrated
Curriculum sequencing	A four-stage sequence moving from critical thinking and independent learning to STEM problem solving, academic communication and review.
STEM lesson development	Selected sessions connected mathematics, science, computing, AI, engineering contexts, reasoning and communication.
Assessment design	Pop quizzes, explanation tasks, homework prompts and presentation scoring were aligned to the purpose of each session.
Data-informed reflection	Learner responses were used to identify understanding, common errors and possible follow-up points.
Professional judgement	The public sample balances evidence of practice with confidentiality, safeguarding and assessment integrity.

What is not published

Institutional logos, school or employer names, staff names, exact dates, internal course branding, raw student data, screenshots, protected assessment materials and identifiable student work are not included.

Selected public wording

I contributed to the development of a mathematics and STEM enrichment sequence for a small cohort of high-attaining international students. My work included STEM lesson development, short assessment tasks, Wayground pop quizzes and review of learner evidence to inform follow-up teaching and feedback.

Evidence should show judgement, not simply activity.